

Advanced Physical Layer Ethernet

Daniel Chidi Chimezie, 1230515, daniel-chidi.chimezie@stud.hshl.de,
Bertrand Mungu Cho, 2220052, bertrand-mungu.cho@stud.hshl.de,
Stephen Uzochi Obuh, 2210044, stephen-uzochi.obuh@stud.hshl.de,
Riwaj Ghimire, 1232171, riwaj.ghimire@stud.hshl.de
Hamm-Lippstadt University of Applied Sciences

Abstract—This seminar paper examines Ethernet-APL (Advanced Physical Layer) as a forward looking communication technology for deployment in explosion-hazardous areas of the process industries. Starting from the limitations of conventional fieldbus solutions such as HART and FOUNDATION Fieldbus, the architecture of Ethernet-APL is explained, which is based on the IEEE 802.3cg-2019 standard (10BASE-T1L) and whose explosion protection is defined by IEC TS 60079-47 through the 2-Wire Intrinsically Safe Ethernet (2-WISE) concept. The paper compares predecessor technologies with Ethernet-APL, analyses application domains, safety concepts and challenges, and evaluates the role of the technology in the context of Industry 4.0 and the Industrial Internet of Things (IIoT). Finally, future perspectives are outlined, including the integration of OPC-UA, Time Sensitive Networking, and cloud-based process data analysis.

Index Terms—Ethernet-APL, 10BASE-T1L, Process Automation, Hazardous Areas, Intrinsic Safety, Industry 4.0, Industrial Internet of Things, OPC-UA

I. INTRODUCTION

The process industries encompassing sectors such as chemicals, petrochemicals, pharmaceuticals, and oil and gas have always placed exceptional demands on industrial communication technology. Field devices such as sensors, transmitters, and actuators frequently operate in potentially explosive atmospheres (hazardous areas), where flammable gases, vapours, or dusts may be present. The operation of electrical equipment in such environments is governed by strict international standards. [1]

While standard Ethernet has been the dominant IT communication technology for decades, there was no means of extending Ethernet connectivity down to field-level devices in explosive atmospheres. Classical solutions such as the analogue 4–20 mA interface or fieldbus protocols like HART, PROFIBUS PA, and FOUNDATION Fieldbus are reaching their limits as data demands from Industry 4.0 and IIoT continue to grow. [2]

In response, Ethernet-APL (Advanced Physical Layer) was developed as a physical layer built on the IEEE standard 802.3cg-2019 (10BASE-T1L), ratified on 7 November 2019, and extended with specific requirements for process automation. The APL Project was initiated in 2018 by twelve leading manufacturers and four standards development organisations, and was successfully completed in 2022. [3], [4]

This seminar paper analyses Ethernet-APL systematically: from its historical context and architecture, through its operating principle and safety techniques, to its significance for Industry 4.0 and future developments.

II. OVERVIEW OF PREDECESSOR TECHNOLOGIES

A. Analogue 4–20 mA Interface

The analogue 4–20 mA current loop has been the industrial standard for transmitting a single process variable per loop since the 1960s. It is characterised by robustness and simple wiring. The minimum current value of 4 mA also enables wire-break detection. A critical disadvantage is its low information density: exactly one measured value is transmitted per cable, and no diagnostic information is possible. The energy levels permissible for Zone 0 also limit the available supply power to approximately 36 mW. [1]

B. HART Protocol

The Highway Addressable Remote Transducer (HART) protocol superimposes a digital FSK (Frequency Shift Keying) signal on the 4–20 mA loop, enabling the transmission of additional diagnostic data and device parameters. However, the effective data rate is only 1.2 kbit/s, which severely limits access to digital information. [4]

C. PROFIBUS PA and FOUNDATION Fieldbus

With the introduction of PROFIBUS PA and FOUNDATION Fieldbus H1 in the 1990s, multiple field devices were connected for the first time over a shared two-wire cable. Both protocols support data rates of 31.25 kbit/s and enable multi-drop topologies. They are specified for explosion protection according to FISCO (Fieldbus Intrinsically Safe Concept). [5] Despite these advances, these technologies are largely vendor-specific and offer neither the bandwidth nor the seamless IT integration required for Industry 4.0 applications. Moreover, data rates of 31.25 kbit/s are wholly insufficient given modern diagnostic and asset management requirements. [2]

D. Comparative Overview

Table I compares conventional fieldbus technologies with Ethernet-APL in terms of communication performance, topology, diagnostic capabilities, and deployment characteristics.

III. FUNDAMENTALS AND ARCHITECTURE OF ETHERNET-APL

A. Normative Basis

Ethernet-APL is based on the IEEE standard 802.3cg-2019 (10BASE-T1L), ratified on 7 November 2019, which defines single-pair Ethernet transmission at 10 Mbit/s over a twisted two wire cable of up to 1000 m. Building on this, the

TABLE I
COMPARISON OF FIELDBUS TECHNOLOGIES

Feature	4–20 mA	HART	PROFIBUS PA	Ethernet-APL
Data Rate	Analogue	1.2 kbit/s	31.25 kbit/s	10 Mbit/s
Cable Length	up to 1000 m	up to 1500 m	up to 1900 m	Trunk: 1000 m, Spur: 200 m
Zone 0 Capable	Yes	Yes	Yes	Yes (2-WISE)
Topology	Point-to-point	Point-to-point	Multi-Drop	Star/Line/Ring
Protocol	Analogue	Proprietary	Proprietary	Ethernet (open)
Power over Wire	Yes (2-wire)	Yes (2-wire)	Yes (2-wire)	Yes (up to 50 W)
Diagnostic Capability	None	Limited	Yes	Comprehensive

technical specification IEC TS 60079-47 specifies explosion protection via 2-WISE. This is supplemented by a Port Profile Specification defining power classes and topology rules, as well as EMC test specifications for the conformity assessment of APL devices. [5], [6]

B. Network Topology: Trunk and Spur

The Ethernet-APL system is structured into two main levels, adapting the trunk-and-spur topology familiar from the fieldbus world: [5]

Trunk (Backbone): Connects APL infrastructure devices (Power Switches, Field Switches) over a maximum of 1000 m in the 2.4 Vp-p operating mode of 10BASE-T1L. The trunk supplies downstream infrastructure components with power (Powered Trunk, up to 57.5 W) or operates without power supply (Unpowered Trunk).

Spur (Stub Line): Connects the Field Switch to individual field devices. The maximum length is 200 m (1.0 Vp-p operating mode), with power limited to intrinsically safe parameters to meet Zone 0 requirements.

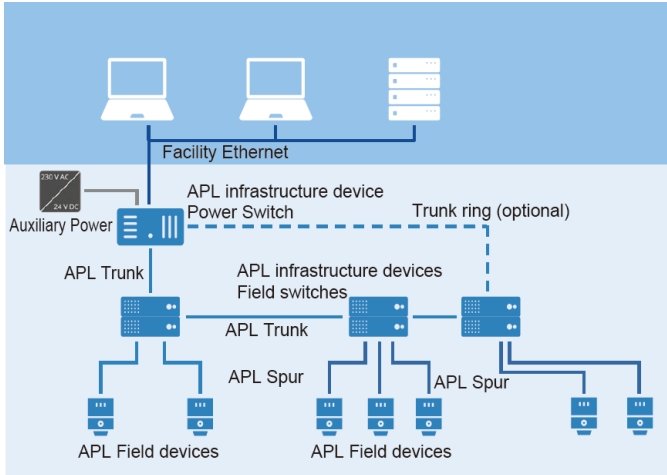


Fig. 1. Typical configuration of an Ethernet-APL system showing trunk and spur topology [4].

C. Infrastructure Devices

The Ethernet-APL system distinguishes three device categories: [5]

Power Switch (PS): Converts standard Ethernet signals to APL signals and feeds power into the trunk.

Field Switch (FS): Distributes signals and power from the

trunk to multiple spur connections. Theoretically, approximately 50 field devices consuming 0.5 W each can be supplied per Field Switch.

Media Converter (MC): Converts standard Ethernet to APL but does not transmit power over the trunk (Unpowered Trunk).

IV. WHY ETHERNET-APL IN HAZARDOUS AREAS?

A. Classification of Hazardous Areas

The IEC 60079 series of standards and the ATEX Directive 2014/34/EU classify hazardous areas according to the frequency with which a dangerous atmosphere is present. Following the methodology described by Bozek (2018) based on IEC 60079-10-1: [1]

Table II summarizes the IEC hazardous area classification and its corresponding NEC division categories.

TABLE II
ZONE CLASSIFICATION ACCORDING TO IEC 60079-10-1

IEC Zone	NEC Division	Description
Zone 0	Div. 1	Explosive atmosphere present continuously or for long periods
Zone 1	Div. 1	Explosive atmosphere likely to occur in normal operation
Zone 2	Div. 2	Explosive atmosphere unlikely in normal operation

B. Challenges for Standard Ethernet

Standard Ethernet (e.g., 100BASE-TX) cannot be deployed directly in hazardous areas for several reasons: [2], [7]

- **Limited Distance:** Cable length is restricted to 100 m, which is insufficient for large process plants.
- **Safety Constraints:** Energy levels exceed those permitted for intrinsically safe installations.
- **Connector Limitations:** RJ45 connectors are not designed for harsh industrial environments and provide no integrated power supply.
- **Electromagnetic Interference:** Switched Ethernet architectures may interfere with sensitive measurement systems.

Ethernet-APL overcomes these challenges through 10BASE-T1L transmission, intrinsically safe power classes, and fieldbus compatible cabling.

V. OPERATING PRINCIPLE AND TRANSMISSION TECHNOLOGY

A. Physical Layer: 10BASE-T1L

IEEE standard 802.3cg-2019 defines 10BASE-T1L as single-pair Ethernet (SPE) with the following core parameters: [6], [7] Signal transmission takes place at 10 Mbit/s in full-duplex mode over a single twisted pair. PAM3 line coding (Pulse Amplitude Modulation with three amplitude levels) achieves efficient spectral utilisation. The standard specifies two signal levels: 2.4 Vp-p for long distances (up to 1000 m) and 1.0 Vp-p for shorter distances up to 200 m, which

suits the intrinsic safety requirements of the spur segment. For the analogue front-end the standard defines insertion loss and return loss limits that are compatible with fieldbus cable type A, allowing existing fieldbus cables to be reused in brownfield plants in many cases.

B. Power Supply and Power Classes

Ethernet-APL transmits both data and power over the same two-wire cable. The standardised power classes are: [5] Table III summarizes the Ethernet-APL power classes and their corresponding explosion-protection categories.

TABLE III
APL POWER CLASSES AND INTRINSIC SAFETY

Class	Segment	Max. Power	Explosion Protection
PS 2-4	Powered Trunk	57.5 W	No IS requirement
FS-A (Ex ia)	Spur	0.54 W	Zone 0 (Ex ia)
FS-B (Ex ic)	Spur	up to 1.84 W	Zone 2 (Ex ic)
FS-C (Ex ec)	Spur	up to 5 W	Zone 2 (Ex ec)

By comparison, a classical 4–20 mA field device in Zone 0 receives only approximately 36 mW. Ethernet-APL already provides approximately 540 mW in the most restrictive class (FS-A), more than 15 times as much, opening the door to more complex field devices with enhanced diagnostics, integrated web servers, or wireless modules.

C. Protocol Independence

As a pure physical layer (OSI Layer 1), Ethernet-APL is protocol-agnostic. All common Industrial Ethernet protocols can be operated over APL: [2], [8] PROFINET, EtherNet/IP, EtherCAT, Modbus TCP, OPC-UA, and proprietary protocols all run unmodified over Ethernet-APL, as the switches act as transparent bridges. Only a bridging device (switch or converter) is required to connect 100BASE-TX and 10BASE-T1L segments.

D. Comparison with Standard Ethernet

Table IV compares the main characteristics of standard Ethernet and Ethernet-APL.

TABLE IV
COMPARISON OF STANDARD ETHERNET AND ETHERNET-APL

Feature	Standard Ethernet	Ethernet-APL
Cable Pairs	4 (Cat5e/6)	1 (Twisted Pair)
Maximum Distance	100 m	1000 m (Trunk)
Speed	100/1000 Mbit/s	10 Mbit/s
Power Supply	PoE (optional)	Integrated (Power-over-DL)
Explosion Protection	Not provided	Zone 0 to Zone 2
Connection	RJ45 / SFP	Fieldbus terminal block
Standard	IEEE 802.3	IEEE 802.3cg + IEC TS 60079-47

VI. SAFETY TECHNIQUES

A. Intrinsic Safety and 2-WISE

Intrinsic safety (Ex i per IEC 60079-11) is the most widely used type of protection for field devices, as it relies solely on the inherent safety of the circuit without mechanical enclosure. The maximum energy entering the atmosphere is kept below the ignition threshold by limiting current, voltage, and capacitance. [1]

2-Wire Intrinsically Safe Ethernet (2-WISE) is the intrinsic safety concept developed specifically for Ethernet-APL and formalised in IEC TS 60079-47. As detailed by Spielmann (2020) in the ODVA conference paper, 2-WISE is based on the proven FISCO concept (Fieldbus Intrinsically Safe Concept) and simplifies its application to Ethernet-APL networks. Users need only verify the 2-WISE marking of the devices and their power class complex energy calculations are not required. ATEX, IECEx, and NEC505 approvals for Zones 0, 1, and 2 as well as NEC500 Div. 1 and Div. 2 are available. [6], [9], [10]

B. Additional Types of Protection

In addition to intrinsic safety, other types of protection may be applied to Ethernet-APL infrastructure devices (Power Switches, Field Switches): Increased Safety (Ex e, IEC 60079-7): Prevents the occurrence of excessively high temperatures and sparks through constructional measures. Flameproof Enclosure (Ex d, IEC 60079-1): Encloses the spark-generating components in a pressure-resistant housing. Encapsulation (Ex m): Embedding in compound material prevents contact with the atmosphere. For spur field devices in Zone 0, however, only intrinsic safety is used.

C. ATEX and IECEx Certification

All equipment deployed in potentially explosive atmospheres in Europe requires ATEX certification per Directive 2014/34/EU. Internationally, the IECEx scheme applies. Ethernet-APL devices undergo extensive conformity tests according to a defined APL Test Framework covering communication, power, and EMC testing in order to ensure interoperability between manufacturers. [1], [9]

D. Cybersecurity

The direct connection of field devices to IT networks via Ethernet-APL significantly increases the attack surface for cyber threats. The IEC 62443 series governs OT security; the survey by Bhamare et al. (2020) provides a comprehensive analysis of these requirements and applicable countermeasures for industrial control systems. Relevant measures for Ethernet-APL networks include: [11] Network segmentation via VLAN configuration in APL Field Switches prevents unauthorised lateral traffic. Role-Based Access Control (RBAC) via OPC-UA restricts access to device parameters. TLS encryption for OPC-UA connections secures data transmission. Monitoring and anomaly detection through Intrusion Detection Systems (IDS) at the network level enable early identification of attacks.

VII. ADVANTAGES AND CHALLENGES

A. Advantages

Ethernet-APL offers significant advantages over classical fieldbus solutions [2], [4]:

- **Speed and bandwidth:** At 10 Mbit/s, Ethernet-APL is more than 300 times faster than HART and 320 times faster than FOUNDATION Fieldbus H1. This enables simultaneous transmission of process values, diagnostic data, device parameters, and even firmware updates over a single connection.
- **Unified communication architecture:** For the first time, a seamless Ethernet connection from the enterprise level down to a field device in Zone 0 is possible without gateways or protocol converters. This reduces complexity and latency.
- **IT/OT convergence:** Field devices can communicate directly with cloud systems, SCADA, and ERP systems. The separation between Operational Technology (OT) and Information Technology (IT) is dissolved without compromising safety.
- **Brownfield compatibility:** The 10BASE-T1L standard is compatible with fieldbus cable type A, allowing existing cabling in installed plants to be partially reused.
- **Vendor independence:** As an open IEEE/IEC standard, Ethernet-APL prevents vendor lock-in and ensures long-term investment security.

B. Challenges

Despite the advantages, significant challenges remain in the adoption of Ethernet-APL [5], [12]:

- **Migration effort:** Transitioning from PROFIBUS PA or HART to Ethernet-APL in brownfield plants requires the replacement of all non-compatible field devices and infrastructure components. In large process plants with thousands of measurement points, this represents a considerable capital and time investment.
- **Device availability:** Although the APL Project was completed in 2022, the range of available APL-native field devices is still limited. The market is growing, but the product variety of classical fieldbus solutions will only be matched in the medium term.
- **Network complexity:** Managing an Ethernet-APL network requires more IT competence than a classical fieldbus system. IT/OT knowledge must be developed within maintenance personnel.
- **Cybersecurity risks:** Direct Ethernet connectivity extending into Zone 0 areas substantially increases the potential attack surface and requires a comprehensive security concept per IEC 62443.
- **Initial investment costs:** APL-capable field devices and infrastructure components are currently more expensive than conventional fieldbus hardware, although falling unit costs are expected as market adoption broadens.

VIII. ROLE IN INDUSTRY 4.0 AND IIOT

A. The Data Gap in the Process Industry

In discrete manufacturing, Industry 4.0 concepts such as digital twins, predictive maintenance, and AI-supported process optimisation are already widespread. In the process industries, however, the low bandwidth of classical fieldbuses has prevented comprehensive data acquisition. Analogue 4–20 mA interfaces transmit only a single process value; HART allows digital diagnostic data, but the data rate is so low that polling a typical plant with thousands of field devices would take hours. [8] Ethernet-APL closes this data gap. With 10 Mbit/s per device, it is possible to transmit not only process values but also secondary process variables, device status, calibration data, NAMUR-compliant diagnostics (NE107), and firmware simultaneously. [2]

B. Digital Twin and Asset Management

The high data bandwidth of Ethernet-APL enables the realisation of Asset Administration Shells (AAS) directly within field devices, as envisioned in the RAMI 4.0 reference architecture model. Each field device can host a web server providing FDI (Field Device Integration)-compliant device descriptions and diagnostic data. This significantly simplifies engineering, commissioning, and predictive maintenance. [8], [13]

C. OPC-UA and Direct Cloud Connectivity

OPC-UA (Unified Architecture) is the open, vendor-neutral standard for semantic data communication in Industry 4.0. ABB and other leading manufacturers have demonstrated that OPC-UA servers can be implemented directly in APL field devices. [13] Combined with OPC-UA TSN (Time-Sensitive Networking), a fully standardised, low-latency communication infrastructure emerges from field level to cloud, linking real-time control loops with simultaneous data availability for higher-level analytics. [14]

D. System Architecture in the Process Plant

Figure 2 illustrates the role of Ethernet-APL within an Industry 4.0 system architecture.

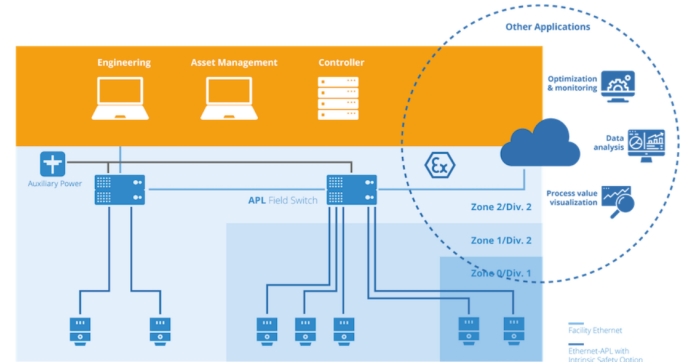


Fig. 2. Ethernet-APL enabling communication from field devices in hazardous areas to engineering, asset management, control, and cloud-based Industry 4.0 applications. [2].

IX. FUTURE TECHNOLOGIES AND DEVELOPMENT TRENDS

A. 100BASE-T1L and Higher Data Rates

Current research is investigating the extension of the SPE standard to 100 Mbit/s over a two-wire cable (100BASE-T1L). This would increase the bandwidth for Ethernet-APL applications by a factor of 10, opening new application possibilities such as high-resolution video signals for pipeline inspection or extensive ML inference directly at the field device. [6]

B. OPC-UA TSN for Real-Time Capability

Time-Sensitive Networking (TSN) per IEEE 802.1Q enables deterministic, low-latency communication in Ethernet networks. Combined with OPC-UA, a vendor-neutral solution emerges that realises closed control loops with cycle times below 1 ms over standardised Ethernet infrastructure. OPC-UA TSN has the potential to replace proprietary Industrial Ethernet protocols and form the foundation for converged IT/OT networks [14].

C. Wireless APL and 5G in Hazardous Areas

Parallel to wired APL systems, wireless technologies for hazardous areas are continuing to be developed. WirelessHART and ISA100.11a have already demonstrated that reliable wireless communication is feasible in industrial and process automation environments, including applications requiring robust monitoring, diagnostics, and control [15].

In parallel, 5G Ultra-Reliable Low-Latency Communication (URLLC) is emerging as a key technology for future industrial networks. URLLC is designed to provide extremely low latency and very high reliability, making it suitable for mission-critical industrial automation, remote operation, and real-time control applications [16].

A hybrid architecture combining wired Ethernet-APL with wireless extensions for hard-to-reach measurement points is increasingly being considered for future process plants. Such architectures can combine the deterministic communication and intrinsic safety benefits of Ethernet-APL with the flexibility and mobility offered by wireless technologies.

D. AI and Edge Computing at the Field Device

The increased power availability in APL field devices (up to 500 mW in Zone 0) combined with the high data bandwidth for the first time enables the integration of AI algorithms directly at the field device. Applications such as local anomaly detection, sensor fusion of multiple measurement principles, and continuous condition monitoring without cloud connectivity are being actively researched. The Asset Administration Shell (AAS) of the Platform Industrie 4.0 forms the semantic framework for device modelling [13].

E. Standardisation Roadmap

The following standardisation activities are currently underway: the Port Profile Specification and the EMC Test Specification are being developed into international IEC standards. An APL Engineering Guideline document supports planners in the design of Ethernet-APL networks. Integration into IEC

61511 (Functional Safety in the Process Industry) is being discussed in order to make APL usable for Safety Instrumented Systems (SIS) as well. [3], [5]

X. CONCLUSION

Ethernet-APL represents a paradigm shift in field-level communication for the process industries. By combining the open IEEE standard 10BASE-T1L, the 2-WISE intrinsic safety concept per IEC TS 60079-47, and protocol independence, Ethernet-APL for the first time delivers unified, high-bandwidth Ethernet communication from the cloud down to a field device in Zone 0. Compared with the predecessor technologies 4–20 mA, HART, and PROFIBUS PA, Ethernet-APL enables not only a dramatic increase in data transfer rates (by a factor of more than 300 compared with HART), but also the full IT/OT convergence that is essential for Industry 4.0 applications such as predictive maintenance, digital twins, and AI-supported process optimisation. The key challenges lie in migrating existing plants (brownfield), cybersecurity requirements, and the necessary development of competence in maintenance personnel. Future developments such as 100BASE-T1L, OPC-UA TSN, and 5G hybrid architectures will further expand the range of applications. For electrical engineers in the field of process automation, a solid understanding of Ethernet-APL is indispensable today, as the technology has evolved from a research project to a production-ready standard that has been commercially available since 2022 and will shape field-level communication for coming decades. [3]

REFERENCES

- [1] A. Bozek, "Application of IEC 60079-10-1 edition 2.0 for hazardous area classification," *IEEE Transactions on Industry Applications*, vol. 54, no. 2, pp. 1881–1888, 2018, open Access via ResearchGate. [Online]. Available: <https://www.researchgate.net/publication/321930442>
- [2] APL Project Consortium, "Ethernet-apl white paper: Bringing the benefits of ethernet to process industries," FieldComm Group; ODVA; OPC Foundation; PROFIBUS & PROFINET International, Tech. Rep., 2019. [Online]. Available: <https://www.ethernet-apl.org>
- [3] ODVA Inc., "APL project successfully completed: Ethernet-APL is here!" Press Release, Frankfurt, Germany, 2022. [Online]. Available: <https://www.odva.org/news/apl-project-successfully-completed-ethernet-apl-is-here/>
- [4] T. Endoh, S. Kuwahara, and S. Takahashi, "Ethernet-APL for evolving field devices and the future of industrial ethernet," *Yokogawa Technical Report English Edition*, vol. 64, no. 2, pp. 91–100, 2021. [Online]. Available: <https://web-material3.yokogawa.com/1/32505/files/rd-te-r06402-005.pdf>
- [5] M. Spielmann, "Ethernet-APL: Advanced physical layer," in *Proceedings of the ODVA Industry Conference*. Frankfurt, Germany: ODVA Inc., 2020, open Access. [Online]. Available: https://www.odva.org/wp-content/uploads/2020/05/2020-ODVA-Conference_Ethernet-APL_Spielmann_Final.pdf
- [6] R. Moyné, J. Iskandar, and D. Tilbury, "Enabling seamless ethernet to the field with 10base-t1l connectivity," Analog Devices Inc., Tech. Rep., 2020. [Online]. Available: <https://www.analog.com/en/resources/technical-articles/enabling-seamless-ethernet-to-field-with-10base-t1l-connectivity.html>
- [7] *IEEE Standard for Ethernet Amendment 5: Physical Layer Specifications and Management Parameters for 10 Mb/s Operation and Associated Power Delivery over a Single Balanced Pair of Conductors*, IEEE Std. IEEE Std 802.3cg-2019, 2019.
- [8] S. Vitturi, C. Zunino, and T. Sauter, "Industrial communication systems and their future challenges: Next-generation ethernet, IIoT, and 5g," *Proceedings of the IEEE*, vol. 107, no. 6, pp. 944–961, 2019.

- [9] R. STAHL AG, "New standards for intrinsically safe ethernet," R. STAHL Technical Blog, Waldenburg, Germany, 2020. [Online]. Available: <https://r-stahl.com/en/global/blog/post-detail/new-standards-for-intrinsically-safe-ethernet/>
- [10] Pepperl+Fuchs, "How to easily implement intrinsic safety with ethernet-APL," Pepperl+Fuchs Technical Blog, Mannheim, Germany, 2023. [Online]. Available: <https://blog.pepperl-fuchs.com/en/2023/implement-intrinsic-safety-with-ethernet-apl/>
- [11] D. Bhamare, M. Zolanvari, A. Erbad, R. Jain, K. Khan, and N. Meskin, "Cybersecurity for industrial control systems: A survey," *Computers & Security*, vol. 89, p. 101677, 2020, open Access via arXiv. [Online]. Available: <https://arxiv.org/pdf/2002.04124>
- [12] OPC Foundation, "Ethernet-APL white paper: A single physical layer for process automation," OPC Foundation, Scottsdale, AZ, USA, 2021. [Online]. Available: <https://opcfoundation.org/markets-collaboration/ethernet-apl/>
- [13] ABB, "ABB brings ethernet-APL with OPC UA to the field," ABB News Center, Zurich, Switzerland, 2021. [Online]. Available: <https://new.abb.com/news/detail/80779/abb-brings-ethernet-apl-with-opc-ua-to-the-field>
- [14] P. Pop, R. Porrmann, E. Scholz, and C. Preschern, "OPC UA TSN: A next-generation network for industry 4.0 and IIoT," *International Journal of Pervasive Computing and Communications*, vol. 19, no. 3, pp. 386–408, 2023.
- [15] T. P. Raptis, A. Passarella, and M. Conti, "A survey on industrial internet with isa100 wireless," *IEEE Access*, vol. 8, pp. 122 177–122 201, 2020.
- [16] 5G Alliance for Connected Industries and Automation (5G-ACIA), "5g for connected industries and automation," 5G-ACIA, Tech. Rep., 2019. [Online]. Available: <https://5g-acia.org>